

Q1-01-A

Give an algorithm to compute x^{77} that only uses multiplications and uses as few multiplications as possible. How many multiplications does your algorithm use?

Q1-02-A

Convert from base 10 to base 2: (a) $(14.75)_{10}$, (b) $(2.7)_{10}$.

Q1-03-A

Convert from base 2 to base 10: (a) $(11110.01)_2$, (b) $(1.0011)_2 \times 2^5$.